

Mini Lithium Battery Safety System (Mini BSS)

Kretto

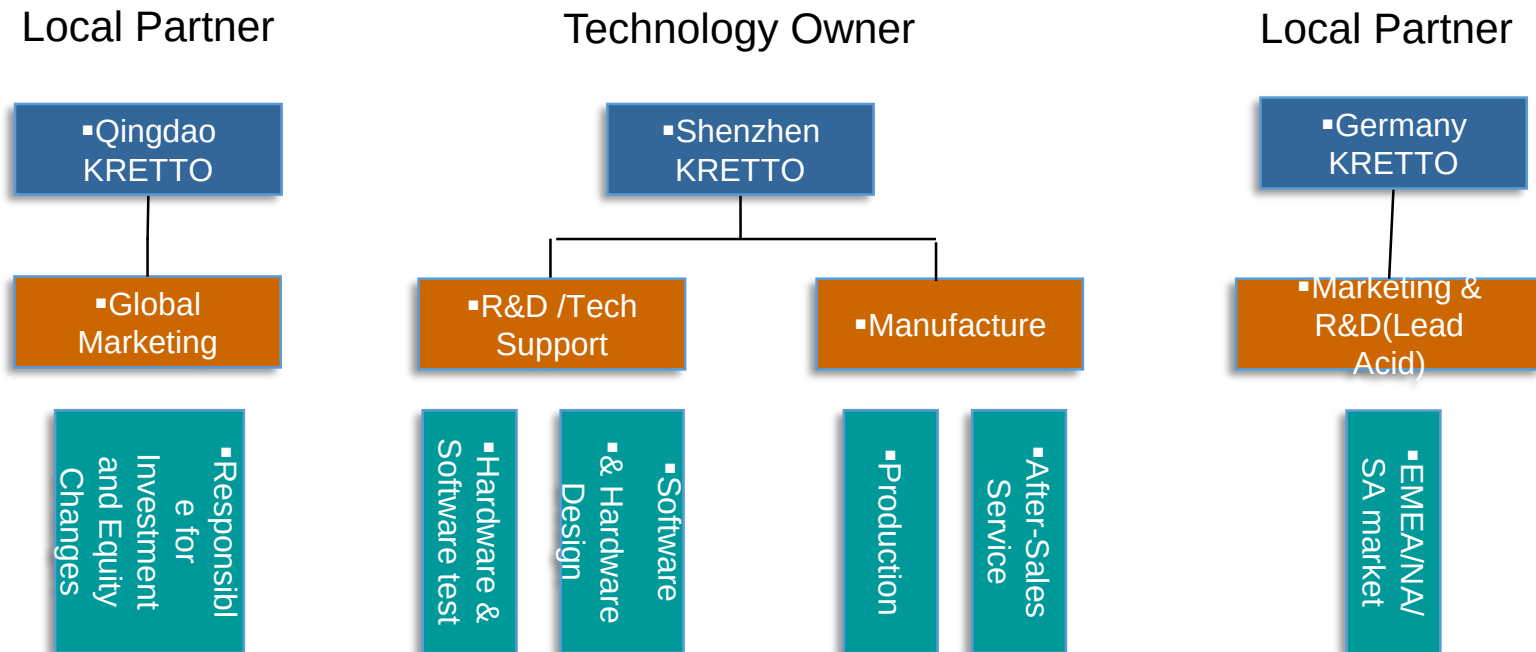
Contents

1. Company Introduction
2. Current Battery Dilemmas
3. Mechanical Sensing Technology for Power Battery Thin Films
4. Low-Cost Intelligent Cell Solutions
5. Research Work Summary
6. Business Model
7. Technical Advancement

Company Introduction

1

▪KRETTO has senior technical personnel with long-term working experience and theoretical knowledge in the fields of batteries, electrical equipment, IT and artificial intelligence, and has formed a network with overseas partners in various countries with global business capabilities.



Dr. Jiang Jiuchun PhD

A leading figure in the power battery management industry, formerly a professor and doctoral advisor at Beijing Jiaotong University, currently serving as Chief Scientist at the Shenzhen Automotive Research Institute, Beijing Institute of Technology.

Recipient of the National High-Level Talent Special Support Program (Ten Thousand Talents Plan),

China Automotive Industry Science and Technology Award (First Prize),

China Machinery Industry Science and Technology Award (Special Prize),

National Science and Technology Progress Award (Second Prize).

Dr. Yan Huiwen PhD

Ph.D. in Power Electronics from Tsinghua University.

Shenzhen Leading Talent.

Expert Reviewer for New Energy Projects.

EMBA from China Europe International Business School (CEIBS).

Former R&D Project Manager at Emerson

Chief Engineer at Kstar

Holds over 20 patents.

Currently an expert committee member and reviewer for the China Power Supply Society.

Dr. Wu Zhiqiang PhD

Senior Engineer, expert registered with the Ministry of Science and Technology, and special expert with the China Association for the Promotion of International Trade.

Long-term engaged in electric vehicle technology research, pioneered the “Three-Network Integration” technology framework and its application. Focuses on battery analysis technology and applications involving mechanism and data fusion.

Former expert member of the Electric Vehicle Standardization Committee in Guangdong Province and Shanghai.

Has led and participated in over 10 national key science and technology projects, and holds more than 10 authorized invention patents.

Dr. Jiang Yan PhD

Senior Engineer with a Doctoral Degree; Visiting Scholar at the University of California, San Diego from 2018 to 2019.

Involved in 10 projects, including National Natural Science Foundation and National Key R&D Program projects.

Published 14 SCI papers and holds 4 authorized invention patents.

Mainly responsible for the development of battery performance evaluation strategies.

Dr. Gao Yang PhD

Senior Engineer, responsible for core algorithm development, with a Doctoral Degree.

Involved in National Natural Science Foundation and National Key R&D Program projects.

Published over 10 SCI and EI indexed papers and holds 4 authorized invention patents.

Primarily responsible for the development of battery safety warning strategies.

Dr. Wei Shaoyuan PhD

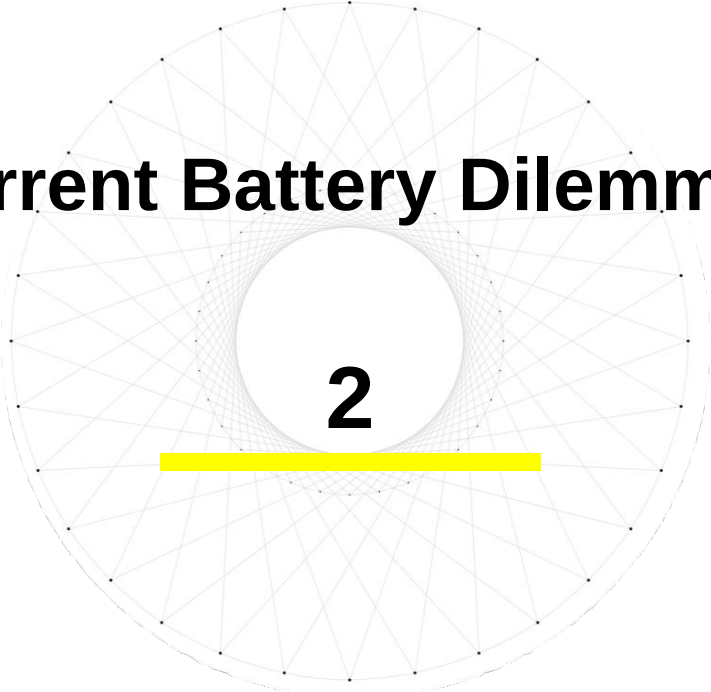
Senior Engineer with a Doctoral Degree; Visiting Scholar at Chalmers University of Technology, Sweden, from 2019 to 2020.

Involved in nearly 10 National Key R&D Program projects and provincial/municipal-level projects.

Published over 10 SCI and EI journal papers and holds 2 authorized invention patents.

Primarily responsible for capacity optimization and energy management strategies.

Current Battery Dilemmas



2

❑ **Battery safety issues** remain a major obstacle to the high-quality development of new energy vehicles.

- **The failure of lithium-ion batteries—particularly the issue of thermal runaway—has become a major focus of industry attention.**
- **In 2023, an average of **six** new energy vehicle fire incidents occurred per day.**
- **In the first half of 2024, there were **10,051** fire incidents involving electric bicycles.**
- **In 2023, the fire incident rate for new energy vehicles was approximately **0.0096%**.**

Fire incidents involving new energy vehicles occur with increasing frequency.

Year	2021	2022	2023	2024 (1H)
Number of new energy vehicles (ten thousand vehicles)	784	1310	2041	2472
Fire data of new energy vehicles (incidents)	> 1500	> 1900	> 1900	—
The fire probability of new energy vehicles (/ten thousand vehicles)	1.85	1.49	0.96	—



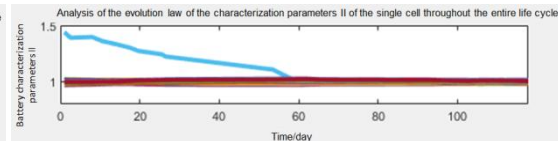
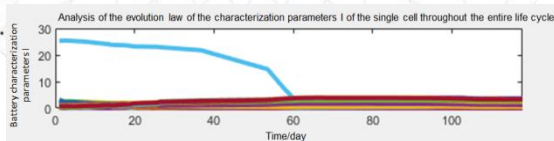
- Over 50% of thermal runaway cases are 'silent failures' with no prior warning. Traditional early warning methods based on electrical and thermal signals are reaching their performance limits.



"Unheralded" thermal runaway



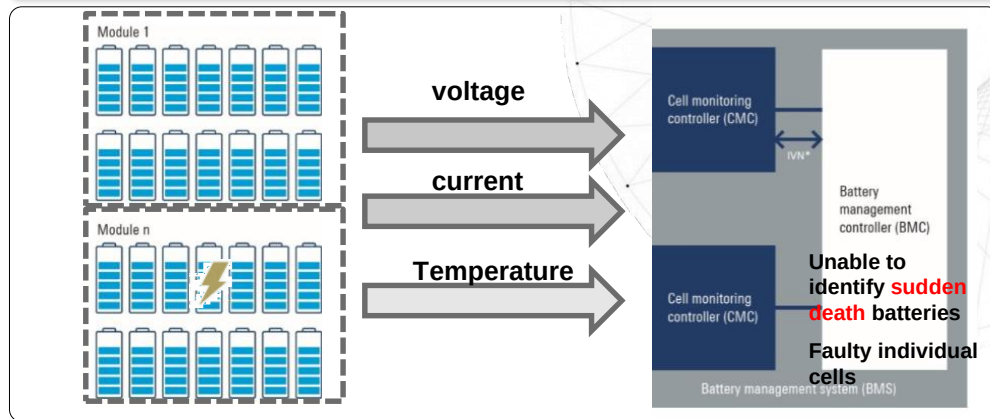
A tens-of-millions-scale operational database of electric vehicles has been established by the team.



Analysis of the long-term evolution of cell parameters over extended time scales.

Early warning of sudden thermal runaway is one of the biggest challenges. Enhancing sensing and monitoring parameters is urgently needed.

- ◆ Traditional BMS estimates battery status by measuring cell voltage, module temperature, and pack current. However, experience in new energy vehicles has shown it cannot provide early warnings for sudden battery failures.
- ◆ Existing BMS systems make it difficult for manufacturers to pinpoint faulty cells. As a result, a quality issue in just a few cells often leads to full-vehicle battery recalls—causing significant costs.



In 2023, there were 15 recalls due to "new energy vehicle batteries".

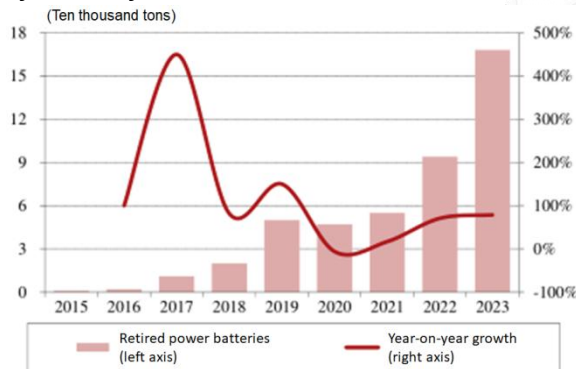
Recall
Announcement
116

Ranking	Recall-related issues	Quantity
1	Engine	18
2	New energy vehicle battery	15
3	Software control / driving assistance	14
4	Body (car windows, doors, sunroof, etc.)	12
5	Airbag	8
6	Seat belt	5
7	Multimedia system	4
8	Seat	3
9	Steering system	3
10	Wiper	3

Traditional BMS has limited signal input, making it unable to predict sudden battery failures or accurately identify faulty cells.

❑ **Illegal battery replacement has become a widespread but unspoken practice in the industry, making it difficult for government agencies to enforce effective oversight.**

- ◆ From 2015 to 2023, the volume of retired power batteries in China increased from 2,000 tons to 168,000 tons. Over the past five years, the compound annual growth rate (CAGR) reached 27.43%. In 2023 alone, 168,000 tons of power batteries were retired nationwide, marking a 78.3% year-on-year increase.



- ◆ After power batteries are retired and recycled, some are illegally refurbished and sold as counterfeit or substandard products, replacing legitimate batteries and re-entering the market.
- ◆ These batteries often have a **short lifespan and unstable performance**, making them prone to safety hazards
- ◆ Currently, there is a lack of effective methods for identifying such batteries.



Recycling and refurbishing of electric vehicle batteries



In February 2024, refurbished batteries caused a fire in Beijing.

There is an urgent need for 'battery passports' to prevent illegal battery replacements and ensure traceability throughout the battery lifecycle

❑ The risks associated with battery storage and transportation are steadily increasing.

- ◆ Lithium-ion batteries are flammable and potentially explosive during storage and transportation, posing **significant risks in the logistics process**.



The *Felicity Ace* cargo ship caught fire and eventually sank while transporting nearly 4,000 vehicles. Industry experts suspect that the fire may have originated from a lithium-ion battery installed in one of the electric vehicles on board.

- ◆ While batteries are closely monitored during usage, they are often left largely unprotected during transportation and storage.
- ◆ Traditional BMS cannot establish a dedicated data storage system for battery cells, nor can it securely store and access full lifecycle data over the long term.



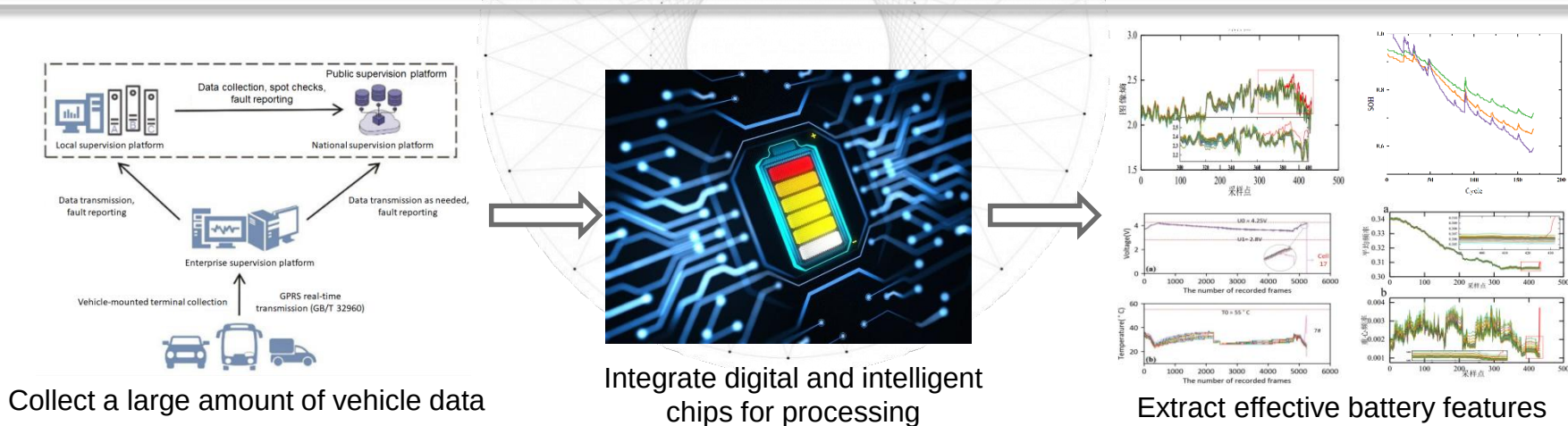
Capture data throughout the battery production process, assign a unique identifier to each battery, and collect full lifecycle data for comprehensive traceability.



There is an urgent need for full lifecycle monitoring and comprehensive battery file data to ensure safety, traceability, and performance optimization.

❑ **Battery data has become a burden rather than an asset**, as it is increasingly difficult to extract meaningful features from massive and unstructured datasets.

- ◆ Through multi-parameter monitoring, **real-time operational data** of the vehicle—such as battery voltage, current, and temperature—can be collected. These parameters enable accurate state estimation and effective safety management of the battery system.
- ◆ The overwhelming volume of battery data has turned it into a negative asset. To extract valuable features, avoid data silos, and reduce the burden on data centers, the adoption of **digital and intelligent chips** is essential.



Data silos are a frequent issue in the industry, highlighting the urgent need for digital and intelligent battery cells to fully unlock the value of massive data.

Mechanical Sensing Technology for Power Battery Thin Films

3

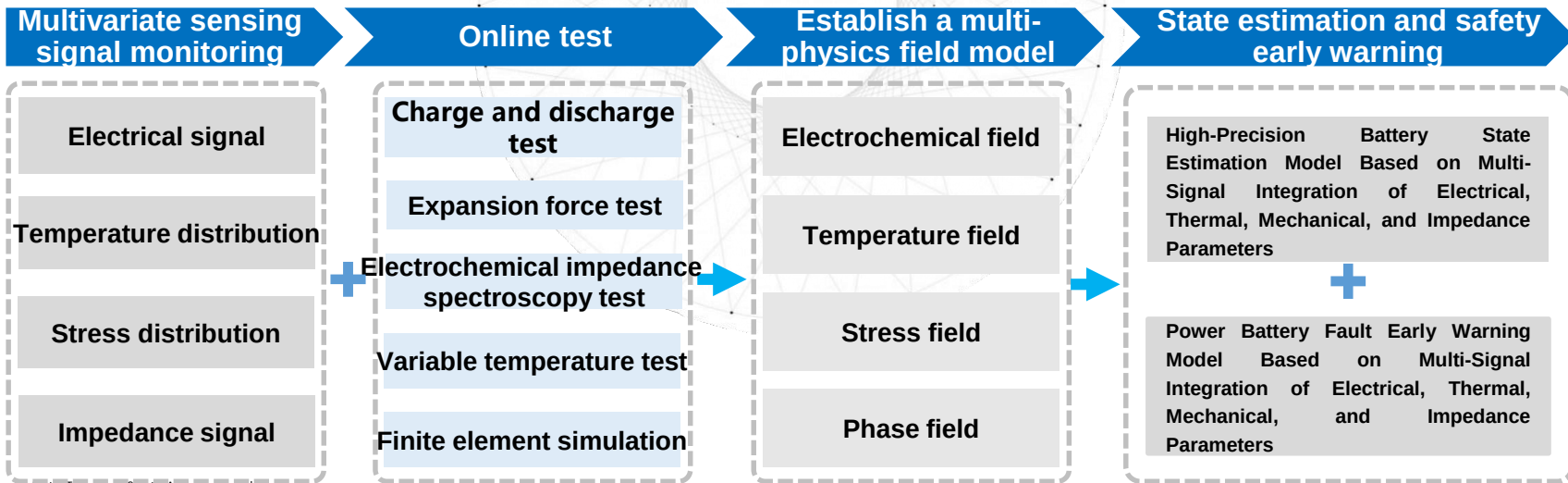


Difficulty

Traditional BMS estimates and manages battery system status based on limited signals such as voltage, current, and temperature. These limited inputs make **it impossible to provide early warnings for sudden-death batteries or to fully identify faulty cells.**

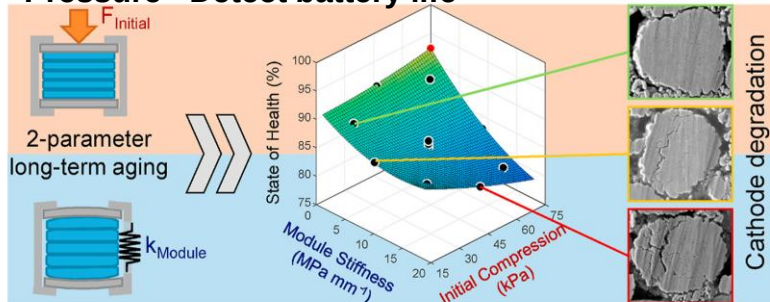
Solution

By applying thin-film mechanical sensing technology to power batteries, mechanical signals are integrated into the state estimation and safety management of battery cells. Combined with multi-parameter monitoring of electro-thermal-mechanical impedance, this approach enhances battery safety through early warning capabilities.



- ❑ The pressure-based battery management approach represents a revolutionary breakthrough in the field of battery applications.

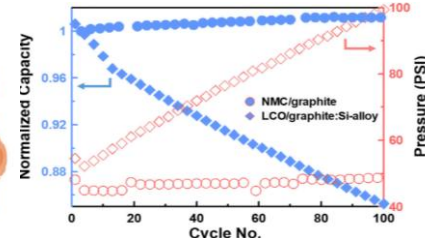
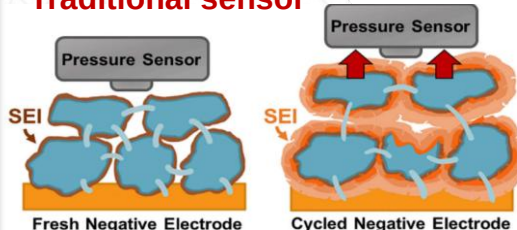
Pressure - Detect battery life



▪ Effects of module stiffness and initial compression on lithium-ion cell aging, Journal of Power Sources, 2021.

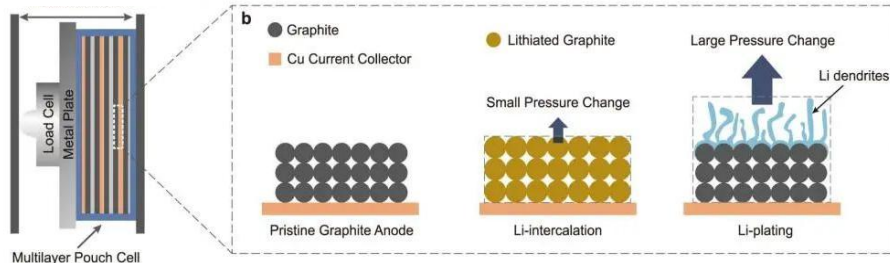
Pressure - Identify battery "sudden death"

Traditional sensor

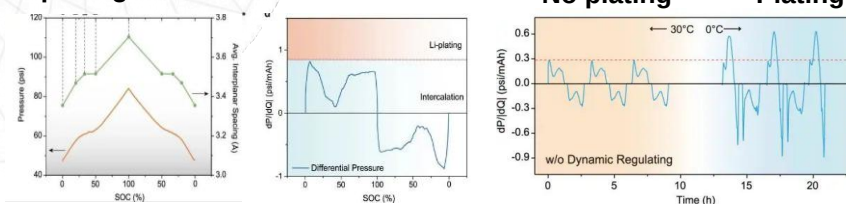


▪ Operando Pressure Measurements Reveal Solid Electrolyte Interphase Growth to Rank Li-Ion Cell Performance, Joule, 2019.

Traditional sensor



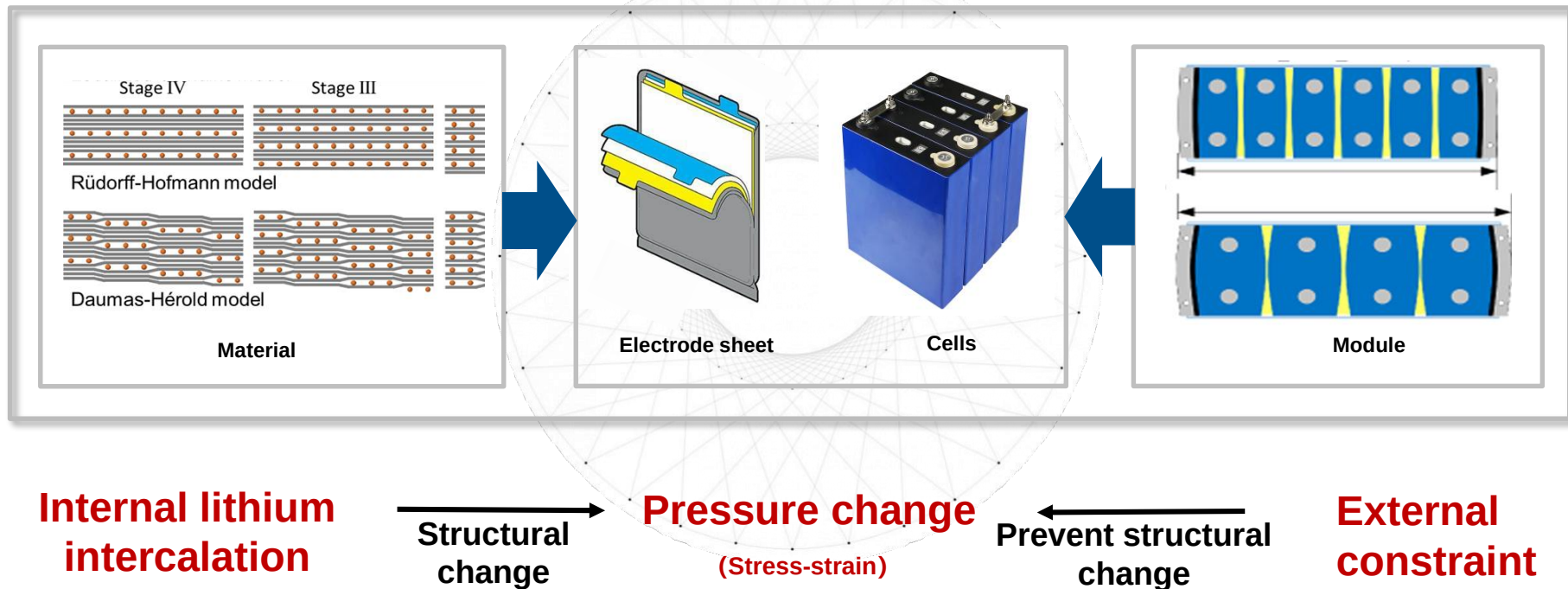
Pressure - Identify lithium plating in batteries



▪ Onboard early detection and mitigation of lithium plating in fast-charging batteries, Nature Communications, 2022.

Pressure sensing has given rise to a range of battery state estimation and fault diagnosis methods.

- From a first-principles perspective, pressure reflects changes in the battery's internal structure—and structural changes, in turn, indicate the battery's state.



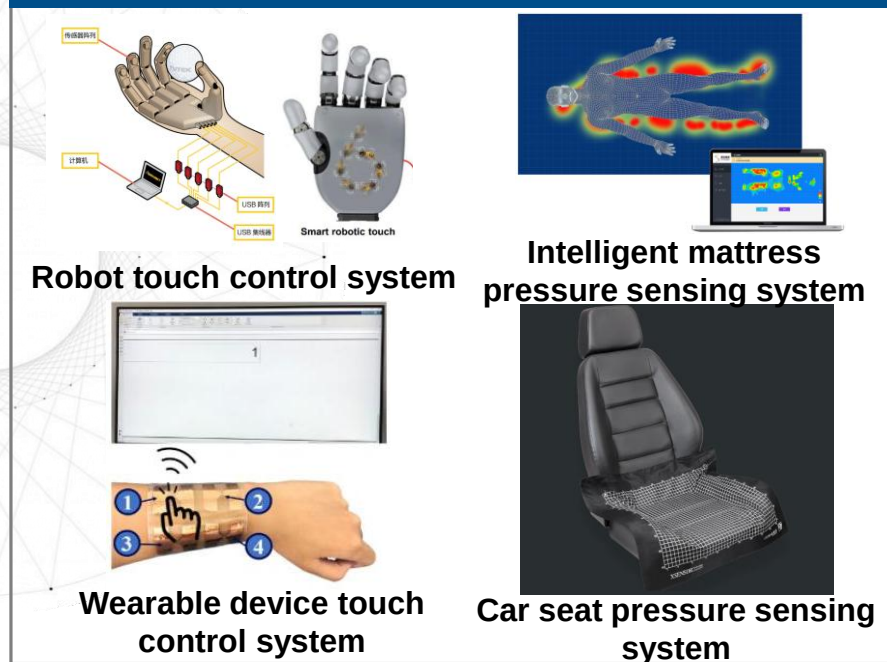
Traditional pressure sensors face spatial limitations in practical applications, restricting their effectiveness in battery monitoring.

- ❑ Flexible tactile (thin-film mechanical) sensors are critical technologies that remain under foreign control, presenting significant challenges for domestic development in China.

35 key technologies that restrict China

1	Lithography machine	19	High-pressure plunger pump
2	Chip	20	Aviation design software
3	Operating system	21	Photoresist
4	Tactile sensor	22	High-pressure common rail system
5	Vacuum evaporation coater	23	Transmission electron microscope
6	Mobile phone RF device	24	Main bearing of roadheader
7	Aero-engine nacelle	25	Microsphere
8	iCLP technology	26	Underwater connector
9	Heavy-duty gas turbine	27	High-end welding power supply
10	Lidar	28	Separator of potassium battery
11	Airworthiness standard	29	Key materials of fuel cell
12	High-end capacitors and resistors	30	Components of medical imaging equipment
13	Core industrial software	31	Database management system
14	ITO target material	32	Epoxy resin
15	Core algorithm	33	Ultra-precision polishing process
16	Aviation steel	34	High-strength stainless steel
17	Milling cutter	35	Scanning electron microscope
18	High-end bearing steel		

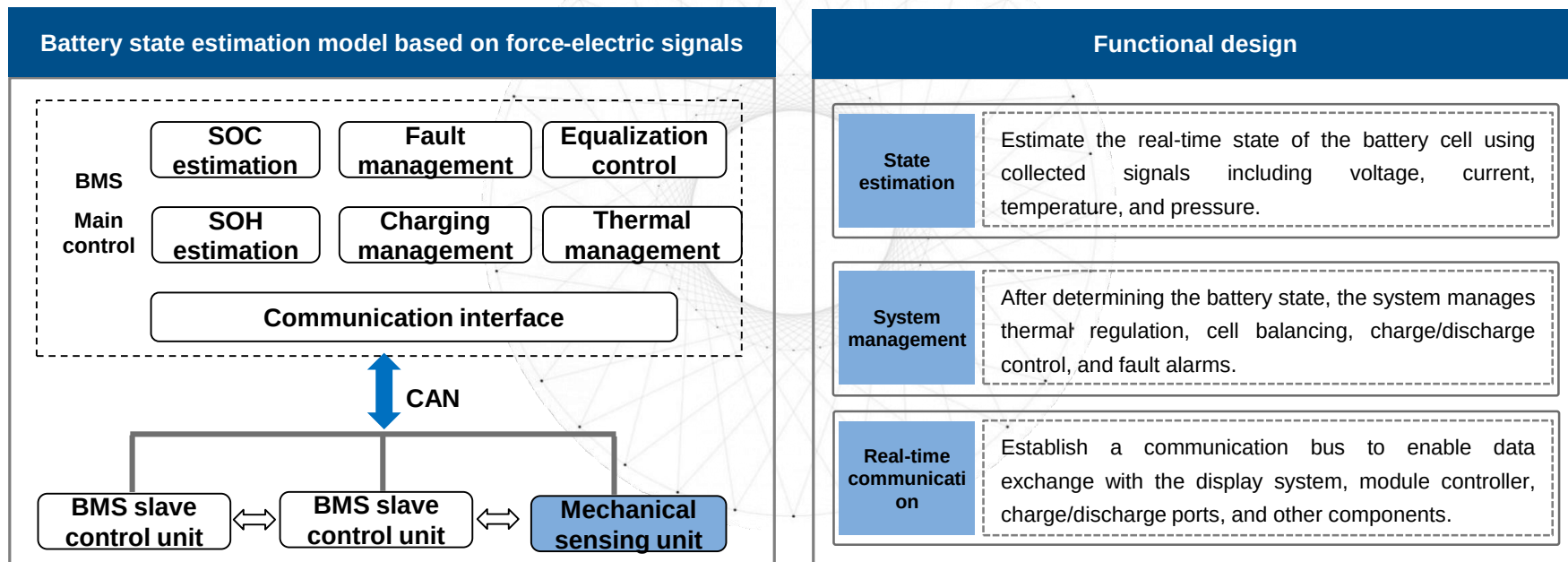
Application fields of thin-film mechanical sensors



There is an urgent need to achieve breakthroughs in flexible thin-film pressure sensing technology and to restore autonomy over key strategic technologies.

❑ BMS Architecture Integrating Mechanical Sensing Units

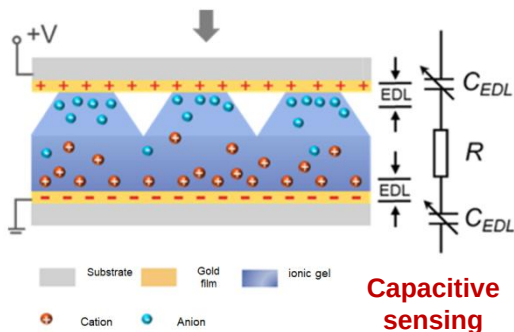
- By integrating thin-film pressure sensors, the system estimates battery SOC (State of Charge) and SOH (State of Health) in real time through the collection and analysis of voltage, current, and in-situ pressure data.



Introduce in-situ mechanical pressure signals to enhance the accuracy of SOX estimation.

- Implemented using fourth-generation dielectric capacitive sensing film, applied for pressure measurement.

Sensing mechanism



Technical advantages

Features high sensitivity, excellent linearity, and minimal creep.

Sensitivity is improved by

120 times

compared to traditional flexible sensors.

Offers a large capacitance value and strong anti-interference performance.

Soft and capable of conforming seamlessly to the skin.

Traditional flexible sensors

	Resistive	Capacitive	Piezoelectric	Dielectric
Sensitivity		✓	✓	✓
Dynamic response		✓	✓	✓
Static response	✓	✓		✓
Noise immunity				✓
Flexibility	✓		✓	✓
Transparency		✓		✓
Manufacturing cost	✓			✓
Power consumption		✓		✓

Core indicators

Fast

Reponse time < 5ms

Resistant

Operating temperature
-20°C ~ 80°C

Accurate

Pressure accuracy
±5% FS

Stable

24-hour drift < 3%

Sensitive

Minimum resolution
< 5g

Excellent

Amplitude attenuation
<3% after 40,000 cycles

The dielectric capacitive sensing film delivers strong performance, high reliability, and significant advantages across key performance indicators.

- ❑ Implements a high-sensitivity pressure distribution sensor and measurement system for battery pressure monitoring applications.

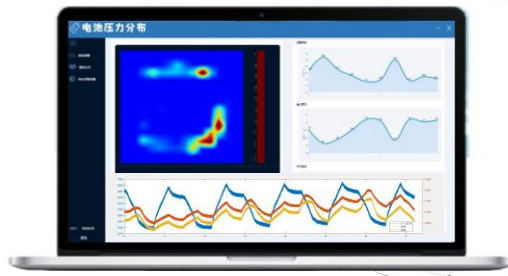
Full set of testing equipment



Pressure
distribution sensor



High-frequency data
acquisition instrument



Special data acquisition
and analysis software

Features and core indicators

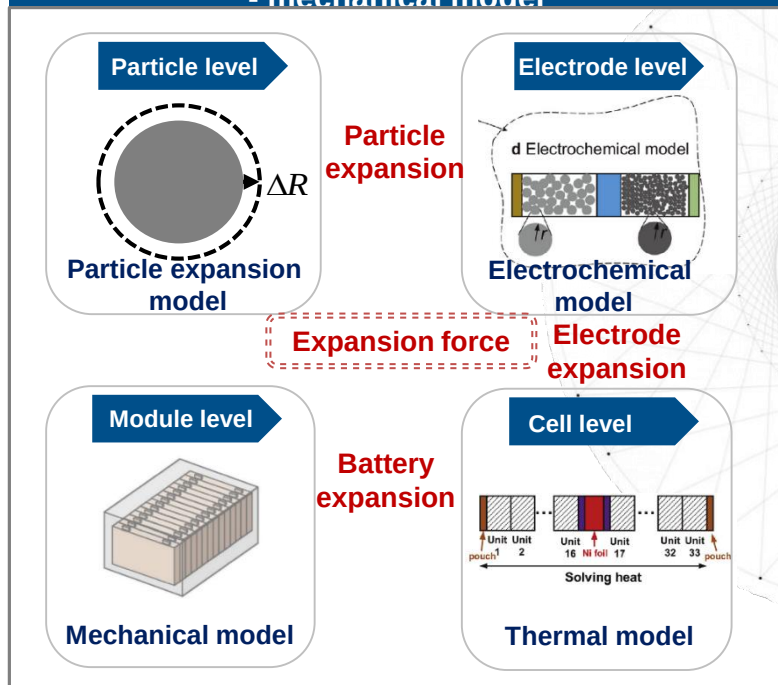
- **Ultra-thin and flexible:** Conforms to the surface of battery cells for real-time pressure distribution monitoring
- **Excellent performance:** High precision, sensitivity, and linearity
- **Wide temperature adaptability:** Operates reliably across varying battery and environmental temperatures
- **Long service life:** Designed for over 10 years of continuous use
- **Customizable design:** Supports integration with battery cells of various sizes and specifications

Sensor performance indicators	Data
Pressure range	0-2Mpa
Number of pressure points	32 (4*8)
Accuracy	<5%
Repeatability	<5%
Non-linearity	<3%
Operating temperature	-20°C-100°C

The pressure distribution sensor is designed for real-time battery pressure monitoring.

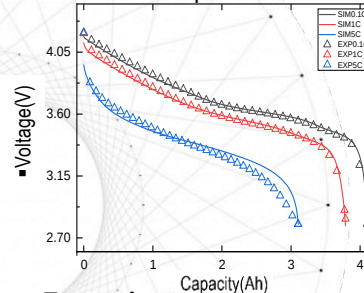
- ❑ Function implemented: Mechanical sensors collect battery pressure information and support battery state estimation

Coupling relationship of electrochemical - mechanical model

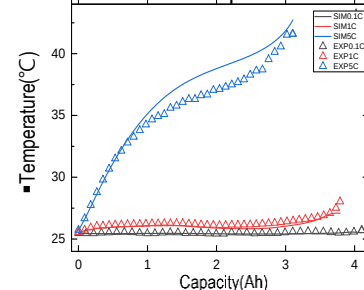


Simulation results of the coupling model

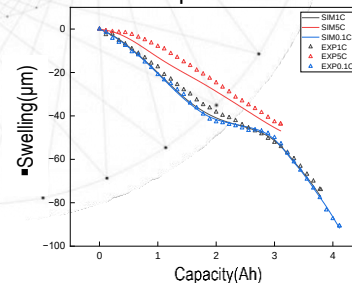
Voltage –
Root Mean Square Error: **62.1mV**



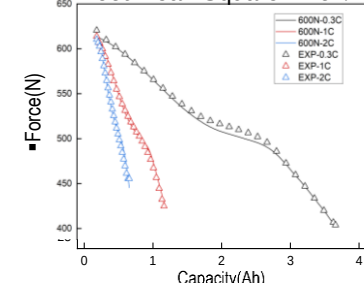
Temperature –
Root Mean Square Error: **2°C**



Expansion –
Root Mean Square Error: **2.62μm**



Pressure –
Root Mean Square Error: **5.04N**



A high-precision battery state estimation model is developed by integrating force-electric signals, utilizing battery expansion force as a key input parameter.

- Function implemented: Establish an electro-thermal-mechanical multi-physics coupling model for the detection of lithium plating in batteries.

Multi-physics field data

Microscopic parameters

Calibration of material
Physical property parameters

▪ D_s , κ , σ , i_0 ...

Calibration of electrode
structure parameters

Areal capacity, Porosity, Tortuosity



Multivariate sensing signals

Electrical signal

Temperature distribution

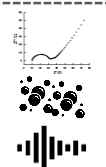
Stress distribution



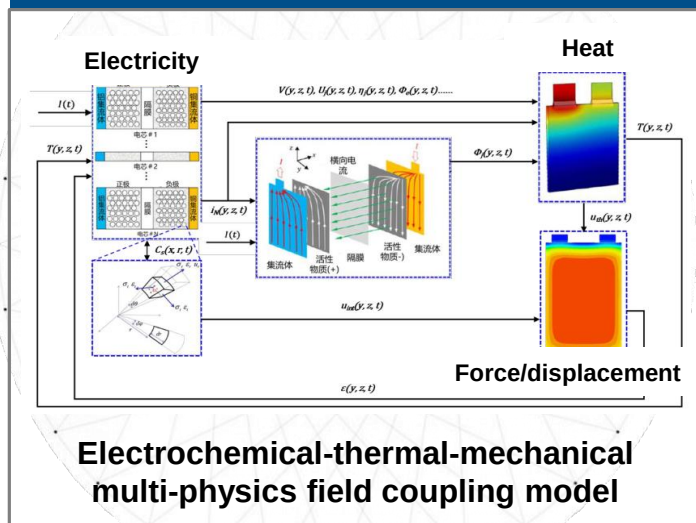
Impedance signal

Battery gas generation

Ultrasonic signal

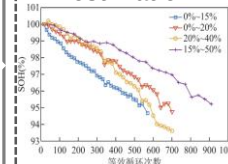


Model coupling

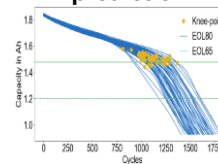


Model application

Capacity estimation



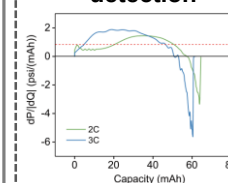
Attenuation prediction



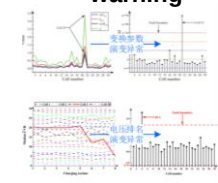
Mechanism
+
Data

Model

Lithium plating detection

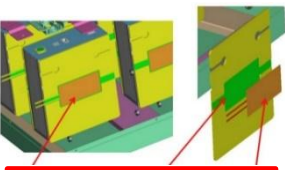
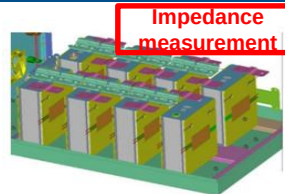


Fault early warning

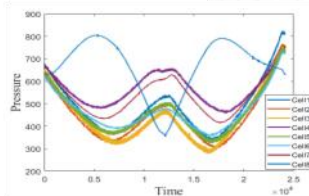


Lithium plating detection in power batteries can be achieved using multivariate information.

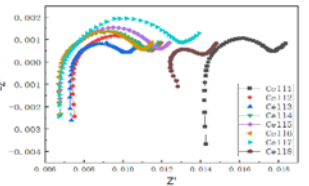
Acquisition of multivariate sensing signals of battery packs



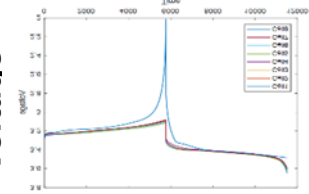
Pressure



Impedance



Voltage



Multivariate sensing analysis data

Statistical characteristic analysis

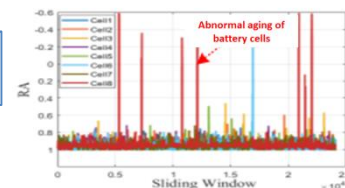
Curve consistency analysis

Differential curve analysis

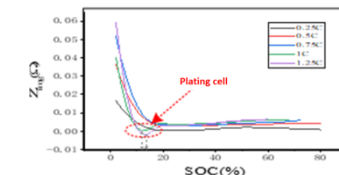
Longitudinal trend analysis

Identification of various faults

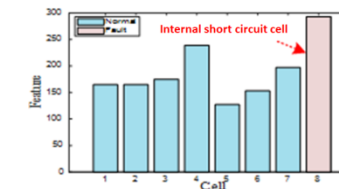
Abnormal aging



Lithium plating

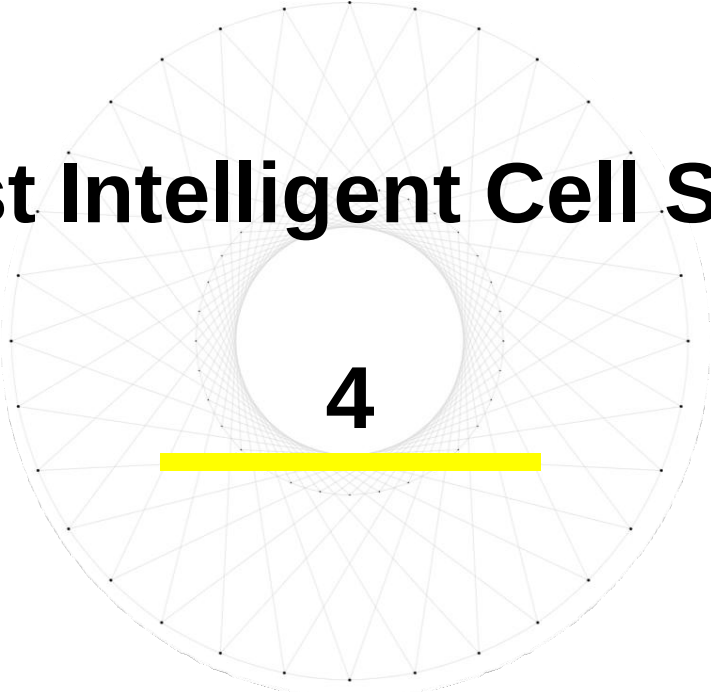


Internal short circuit



The fault early warning model based on multivariate sensing signals can effectively identify various faults and ensure the safe operation of batteries

Low-Cost Intelligent Cell Solutions



4



Difficulty

The challenge of full lifecycle monitoring for battery cells.

- ❑ **Illegal battery replacements have become a hidden norm in the industry**, and the management of the battery supply chain and recycling operations remains imperfect.
- ❑ **The risks associated with battery storage and transportation are continuously increasing**, with these processes often left unprotected.
- ❑ **Battery data has turned into a liability**, and there is an urgent need for digital and intelligent battery technologies to unlock the full potential of massive data.



Solution

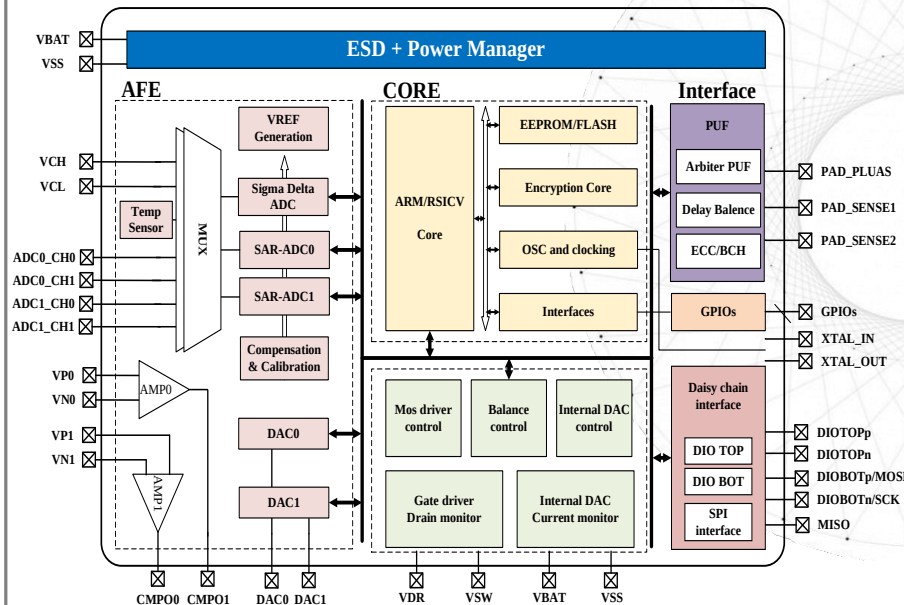
Develop an integrated chip for battery passports and multivariate sensing

- ❑ **Develop a new type of chip-based anti-counterfeiting label technology** to create an unclonable battery identity label that is securely bound to the battery cell.
- ❑ **Develop data encryption and security authentication technology for battery passports**, enabling the secure storage and transmission of battery passport information via the chip.
- ❑ **Develop 'electro-thermal-mechanical' integrated data acquisition and processing technology** for battery cells, allowing the chip to efficiently process distributed data from battery cells and reduce the load on data centers.

Enable high-intensity anti-counterfeiting authentication, secure storage and transportation, and high-precision data processing throughout the full lifecycle of battery cells

Integrated Chip for Battery Passport and Multivariate Sensing – Architecture and Key Functions

Chip architecture



Overall function

Sensing performance:

Battery voltage range: 2 - 5 V

Voltage measurement accuracy: ± 2 mV

Temperature sensor accuracy: ± 2.5 K

Stress detection accuracy: 1Mp

Passport performance:

PUF anti-counterfeiting label (256 bits)

Built-in passport information storage space 64 Kb

Supports external passport information Flash storage

Supports encrypted storage

Battery cell control:

Single battery cell management

Integrated ARM/RISCV core

Supports internal and external balancing of the chip

Two independent voltage measurement channels

Communication interface:

Maximum number of ICs for daisy chain communication: 252

Differential daisy chain communication speed: 1 Mbps

Configurable SPI as a bridge

The world's first chip integrating temperature, voltage, stress monitoring, and battery passport functions.

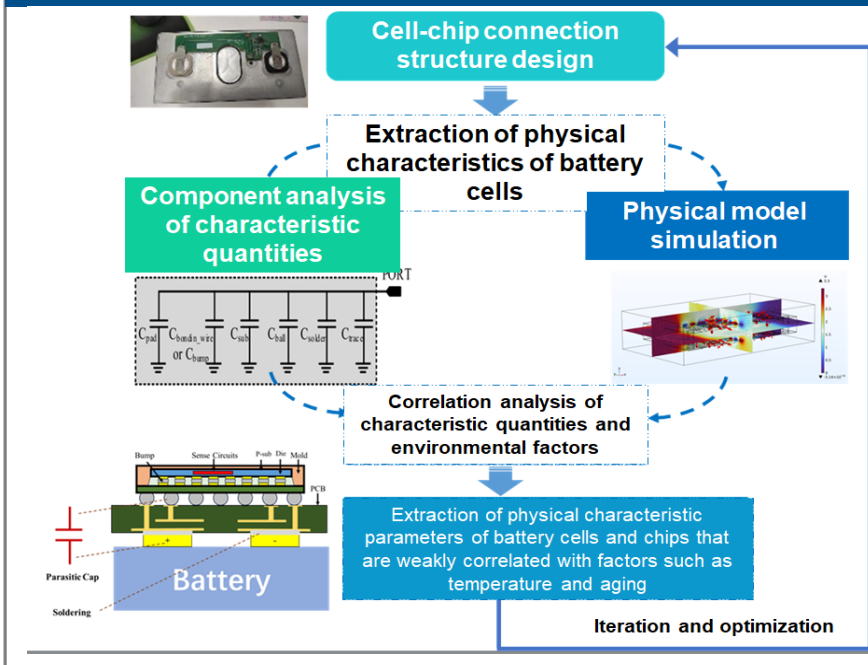
❑ Integrated Chip for Battery Passport and Multivariate Sensing – Benchmarking Against International Standards

Chip model		DNB1100A	DK8x02	BAT100x
Single battery cell management		√	√	√
Front-end AFE parameters (working conditions @VDD = 3.3 V, Temp = 25°C)	Battery voltage range	1.9 to 5.5 V	1.4 to 5.0 V	2.0 to 5.0 V
	Voltage measurement accuracy	±5 mV	±0.6 mV	±2 mV
	Temperature measurement accuracy	±2.5 °C	± 2 °C	±2.5 °C
	External temperature sensing	x	√	√
	Stress detection accuracy	x	x	1 Mpa
	External balancing	√	x	√
	Internal balancing current	200 mA	200 mA	x
Passport function	PUF anti-counterfeiting label	x	x	√
	Passport storage space	x	16 Kb	64 Kb
	Support encrypted storage	x	x	√
Communication mode		Differential daisy chain communication	Wireless communication	Differential daisy chain communication
Working range		-40 °C to +105 °C	-40 °C to +105 °C	-40 °C to +105 °C

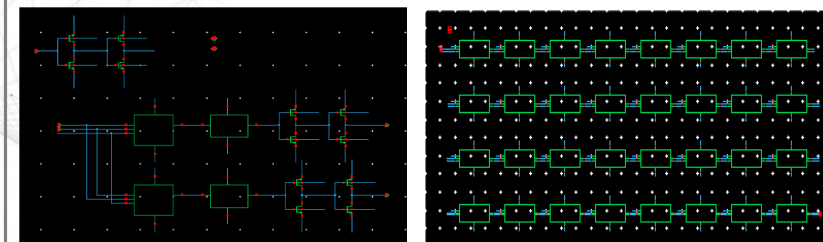
The world's first high-precision electro-thermal-mechanical integrated monitoring chip and the first integrated chip combining battery passport and multivariate sensing."

❑ Integrated Chip for Battery Passport and Multivariate Sensing – Battery Passport Anti-Counterfeiting Label

Extraction of physical characteristics of battery cells

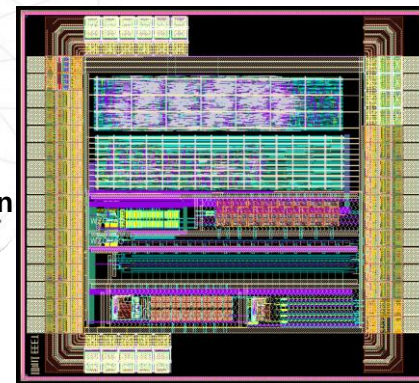


Tapeout verification



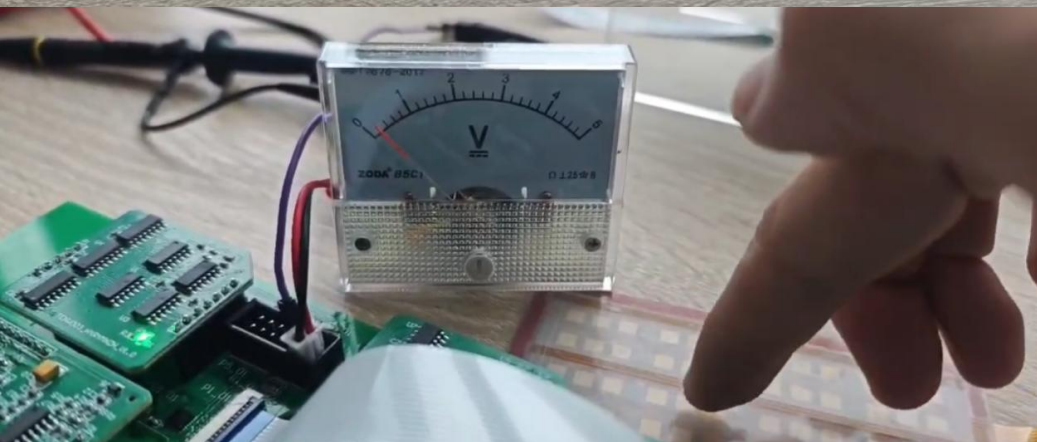
Circuit configuration

Chip layout



Research on chip anti-counterfeiting technology to develop unclonable labels and establish a secure binding between battery cells and chips

Integrated Chip for Battery Passport and Multivariate Sensing – Chip Verification System"

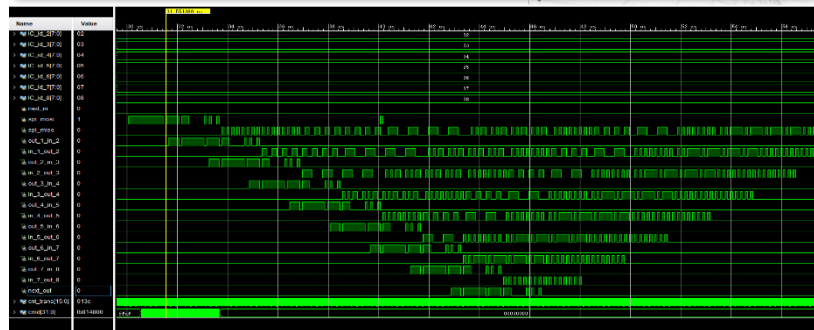
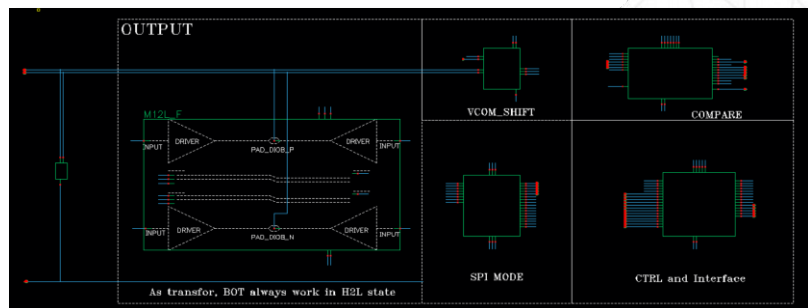


Demonstration based on the 280020 minimum functional system

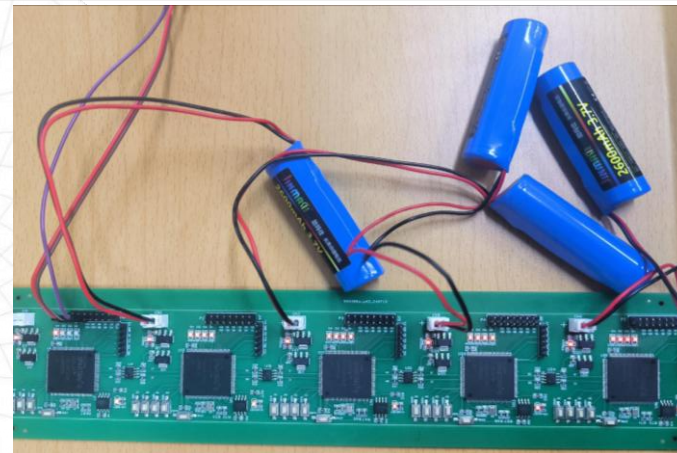
Develop microchips to enable high-intensity anti-counterfeiting certification for individual battery cells, addressing the issue of illegal battery replacements.

❑ Integrated Chip for Battery Passport and Multivariate Sensing – High-Speed Secure Daisy Chain Communication Module

Circuit design



FPGA verification result

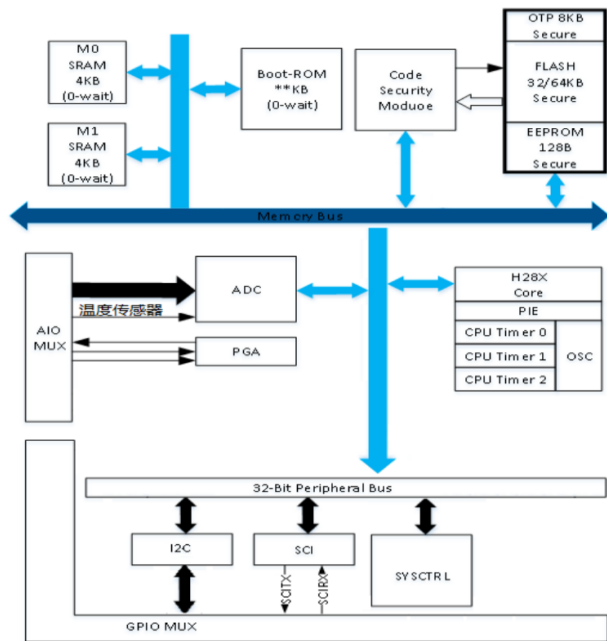


- Complete the analysis of transmission code and communication protocol for daisy chain communication.
- Finish writing and simulating the daisy chain communication code.
- Conduct FPGA board-level verification of the daisy chain communication code, achieving a communication rate of 1 Mbps.

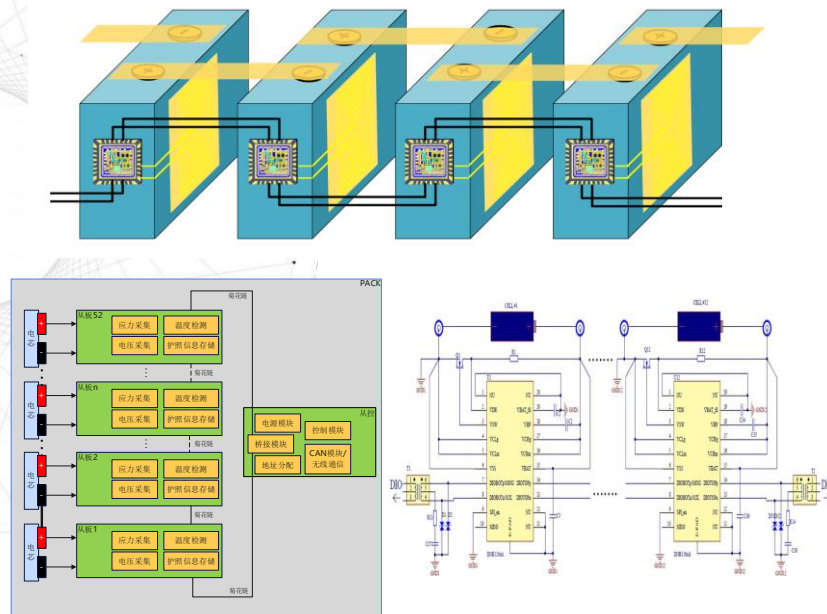
The daisy chain communication interface integrates digital fingerprint keys, hash operations, and lightweight encryption algorithms to ensure encrypted data storage and transmission.

❑ Integrated Chip for Battery Passport and Multivariate Sensing – Data Encryption and Security Certification

Data security storage circuit

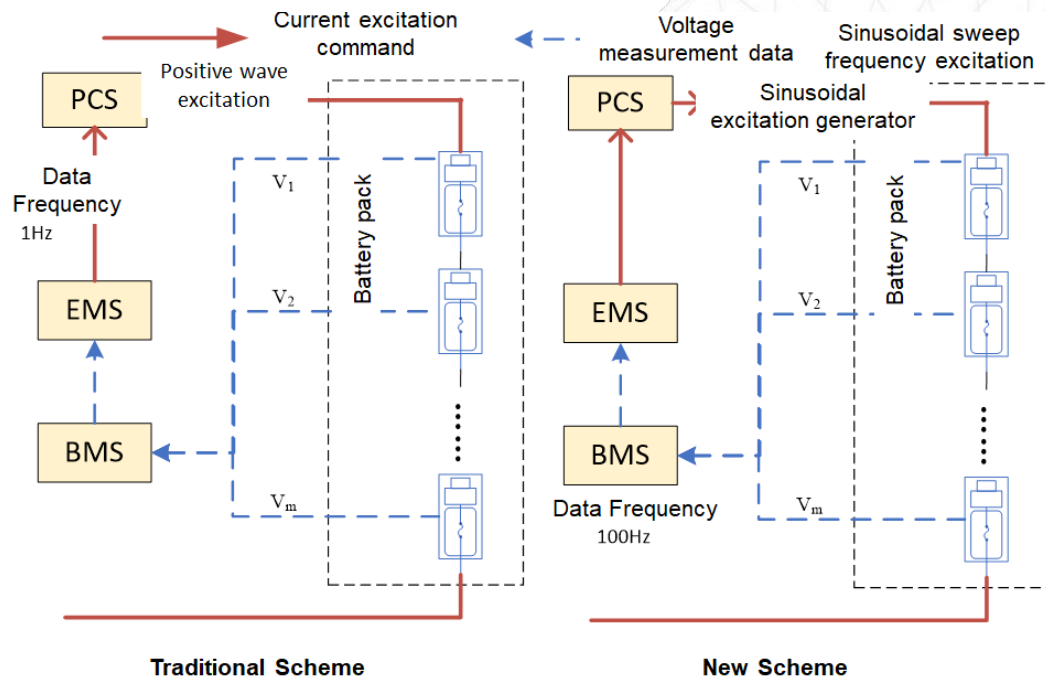


Application examples



Ensure the safe storage, transportation, and use of battery cells throughout their lifecycle, reducing risks associated with battery storage and transportation.

❑ Integrated Chip for Battery Passport and Multivariate Sensing – Battery System Impedance Spectrum Detection Solution

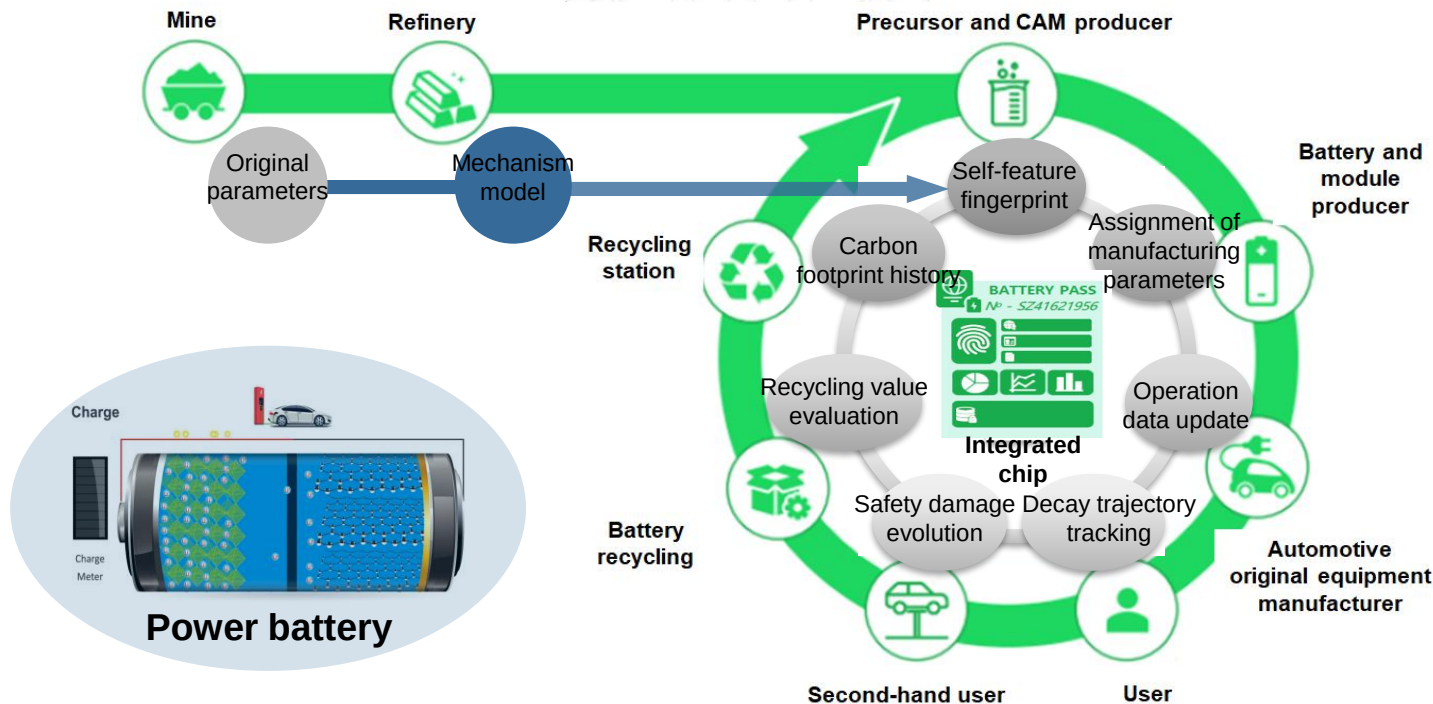


Scheme comparison

Traditional Scheme	New Scheme
Square Wave Excitation	Sinusoidal Excitation
Requires FFT Transformation	No Transformation Needed
Data Frequency $\leq 1\text{Hz}$	Data Frequency 100Hz
Relies on Edge Terminal/ Cloud Computing for Impedance	Impedance Calculated by Chip

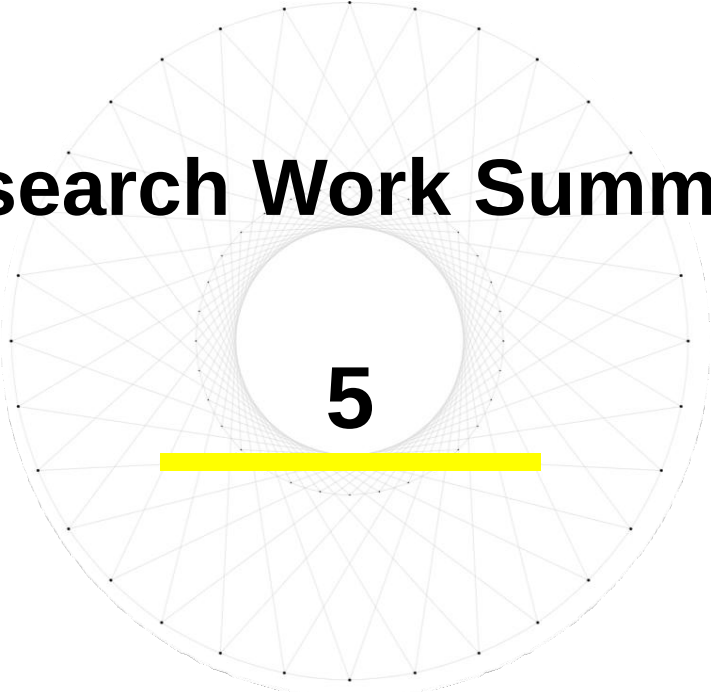
The chip calculates impedance, expanding the frequency range of the impedance spectrum, with local computation and no need for additional resources.

Integrated Chip for Battery Passport and Multivariate Sensing – Battery Feature Extraction and Data Processing



Use an MCU to perform battery data feature extraction, addressing distributed data processing issues and alleviating the load on the data center.

Research Work Summary



5

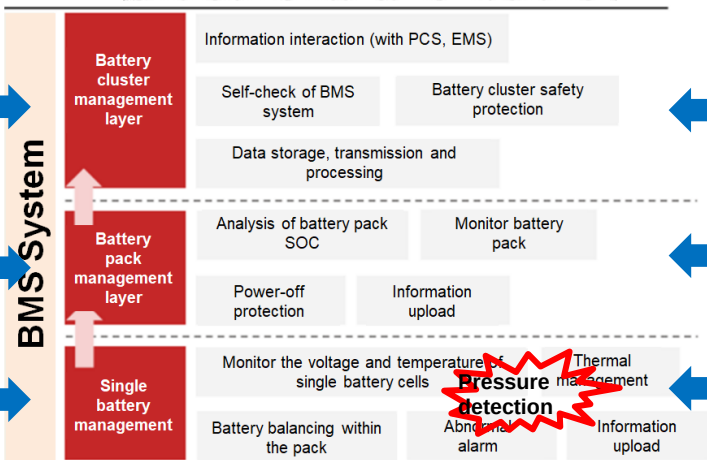
❑ MCU 'One Chip, One Management' – New BMS System Powered by the Next-Generation Cell Management Chip (CMU)

Composition and **Function** Division
of the Existing BMS System

Master control - Data storage,
transmission and processing

Main control - Data summary and
transmission

Slave control - Data **acquisition** and
transmission



Composition and **Computing Power** Division
of the New Generation BMS System

Master control ---- Data storage, transmission
and processing

Main control - Data summary and
transmission

Slave control - Data acquisition, **feature
extraction, edge AI computing** and
transmission

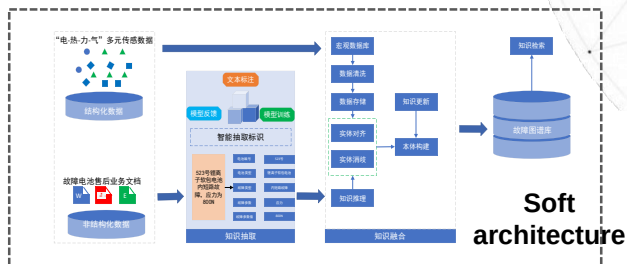
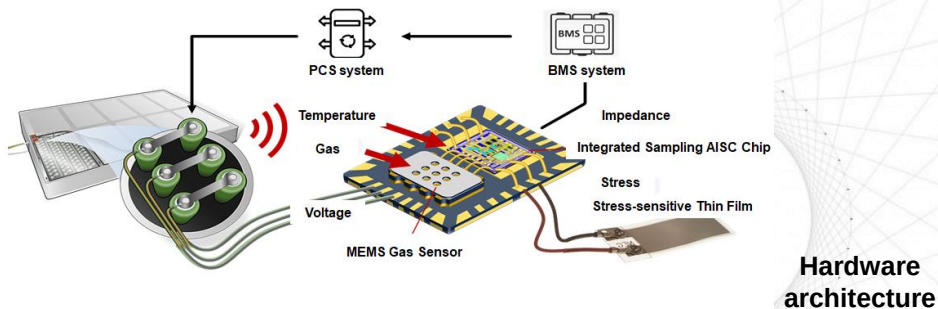
**Pressure
detection**

- Data feature extraction and storage reduce the pressure on data centers, lowering storage and transmission costs across the battery cell management network.
- The distributed computing approach, utilizing edge AI, decreases the demand for data center computing power, storage, and maintenance, enabling cost-effective large-scale deployments.
- More intelligent management of individual battery cells ensures comprehensive safety across their entire lifecycle, providing all-around monitoring and control during production, transportation, storage, usage, and recycling.

The new generation CMU – Advanced sensing technology, greater flexibility, supporting data feature extraction and edge AI computing.

❑ Lithium Battery Safety System Architecture – Integrating Mechanical Monitoring into Cell State Estimation and Safety Management

Battery state estimation model based on multi-signals
of force + electricity - heat - impedance



Core technologies of the entire set
of Mini BSS

Precise early warning of faults
through cloud-edge collaboration

Large model of the cloud-based
fault knowledge graph library

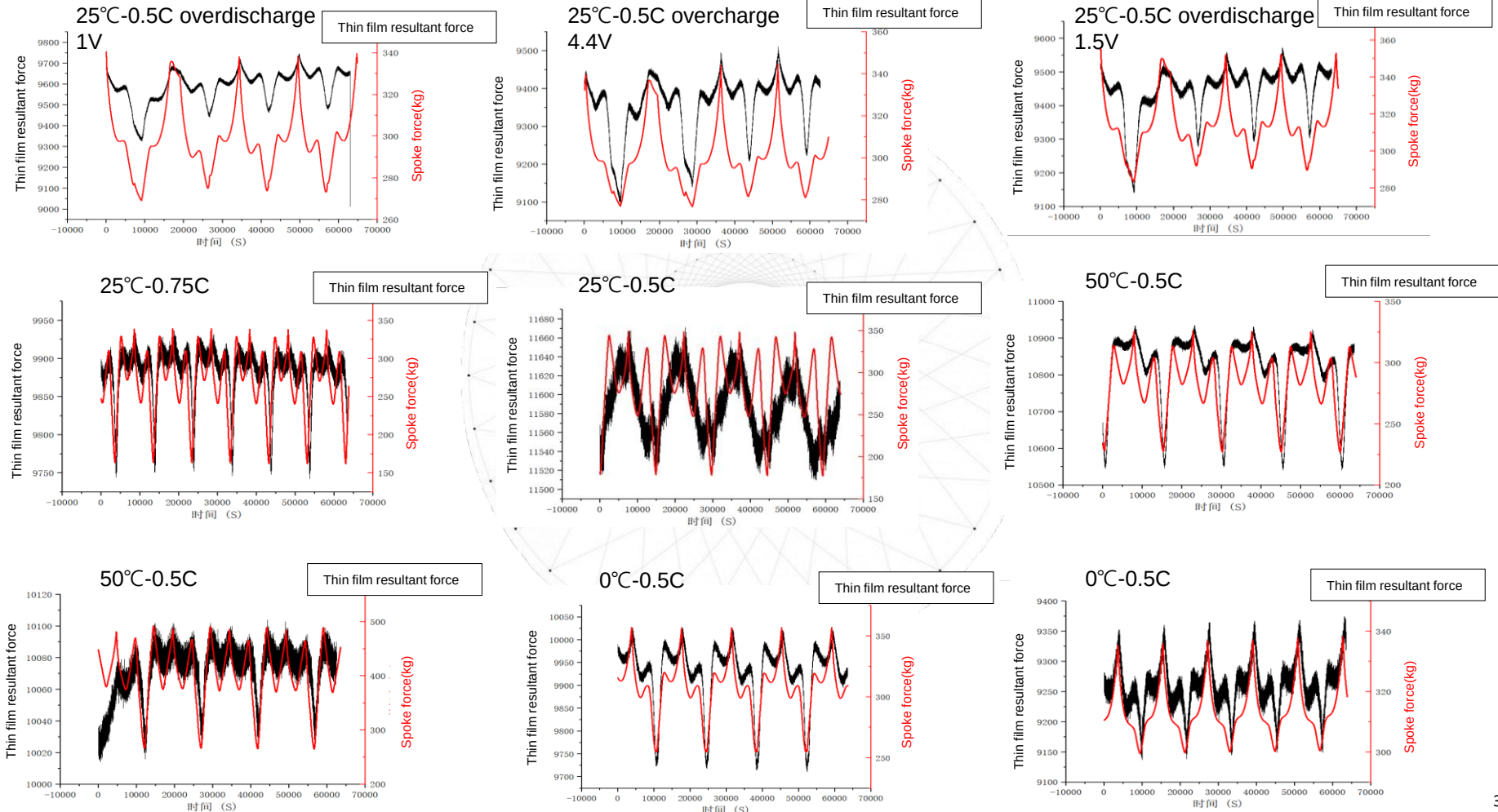
Precise state estimation and
system management of BMS

High-speed communication,
data protection and battery passport

Thin-film sensor + multi-sensor
acquisition chip

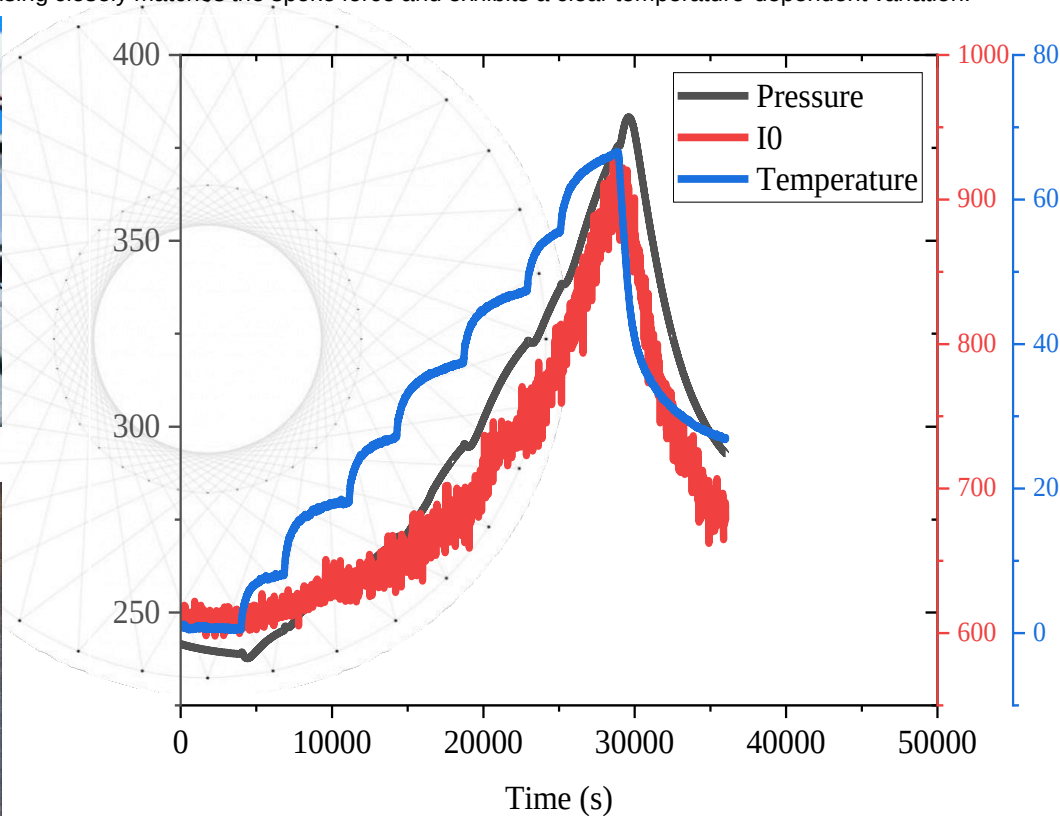
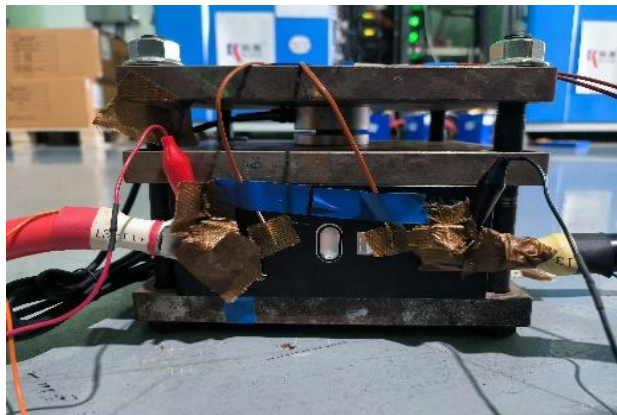
Lithium Battery Sensing Chip – New BMS Management with a Complete System Service Framework for Cloud-Edge Collaborative Safety Early Warning

§Spoke force, thin film variable temperature charge and discharge experiment



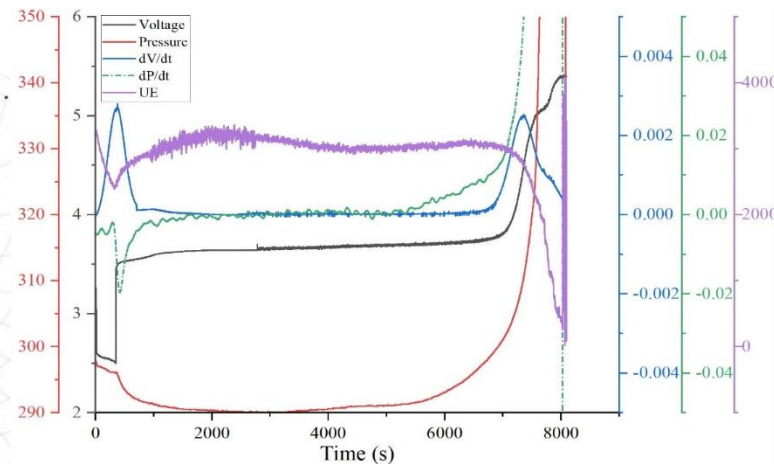
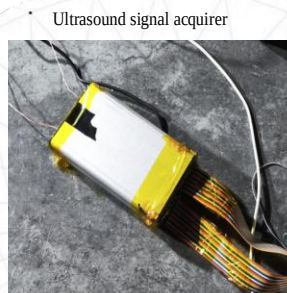
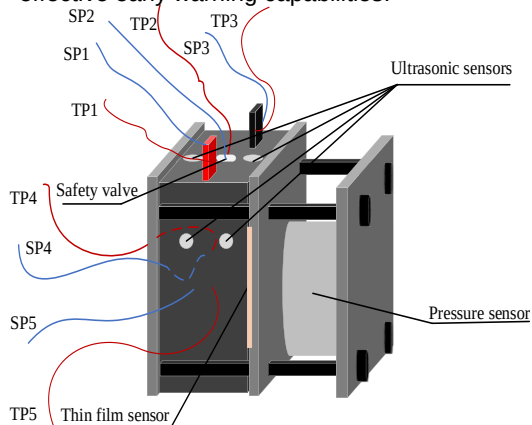
Standing experiment of spoke force and thin-film sensing at various temperatures.

In the non-charging static state (0 - 70°C), the thin-film sensing closely matches the spoke force and exhibits a clear temperature-dependent variation.



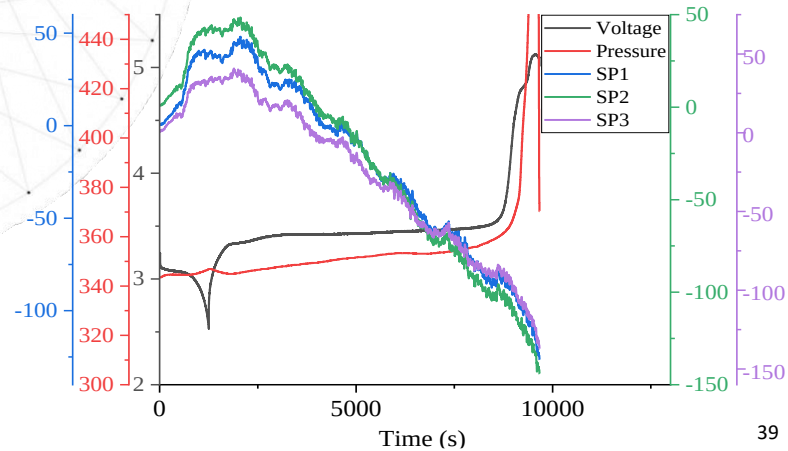
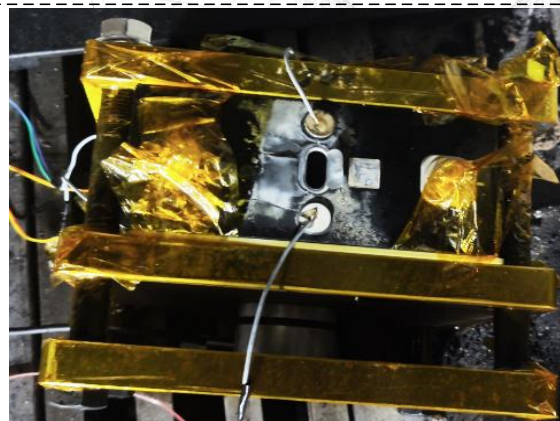
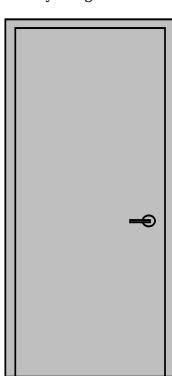
Thermal runaway experiment using spoke force, strain, and ultrasonic sensing.

With a preload of 300 kgf, the pressure signal provides an early warning 3,000 seconds before the battery safety valve opens. Strain and ultrasonic signals also demonstrate effective early warning capabilities.

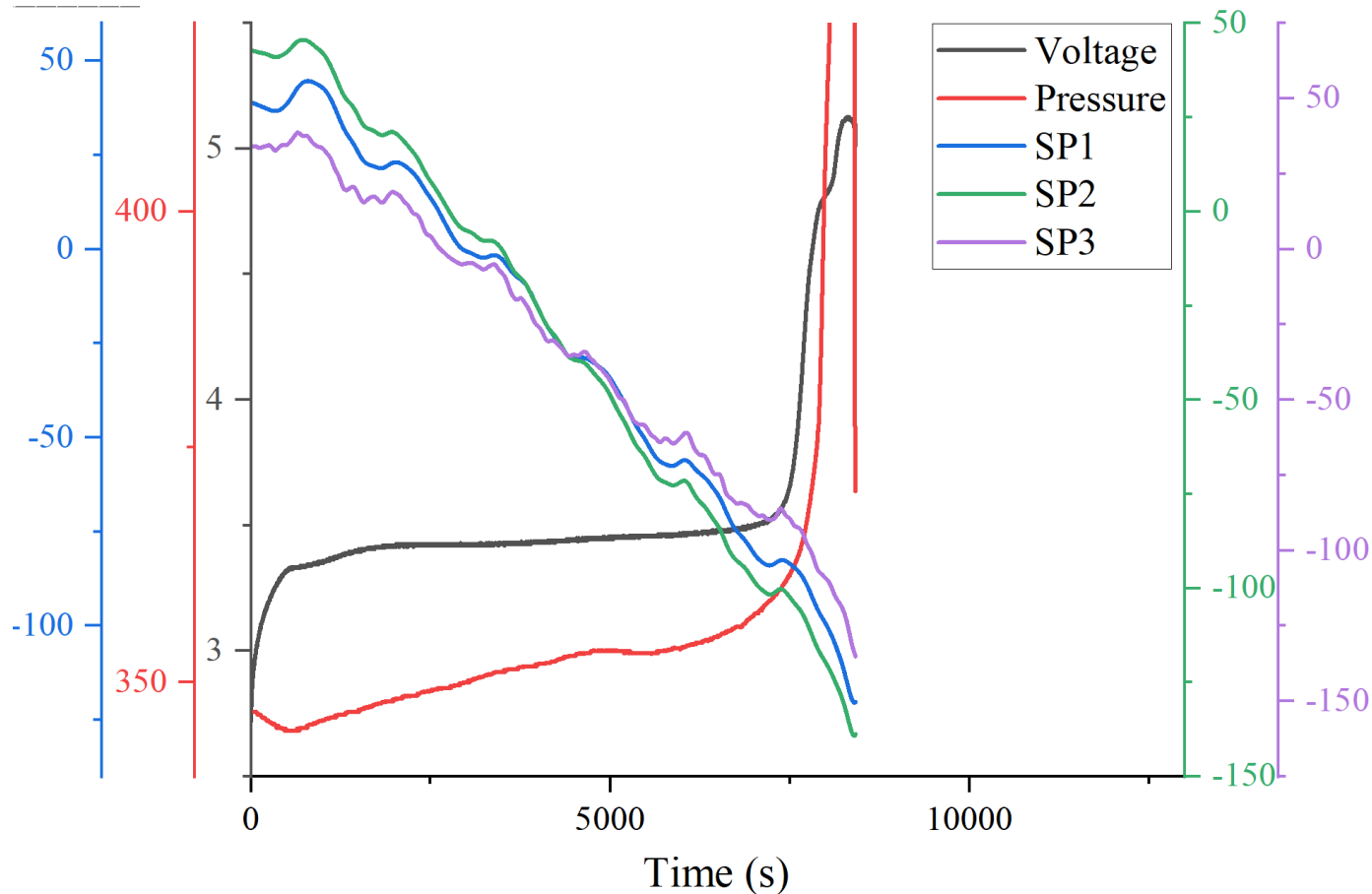


Battery testing device

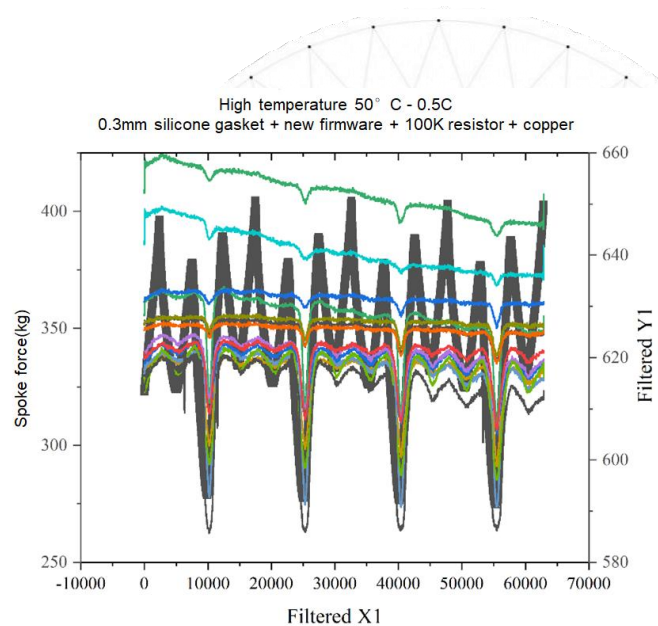
Safety safeguards room



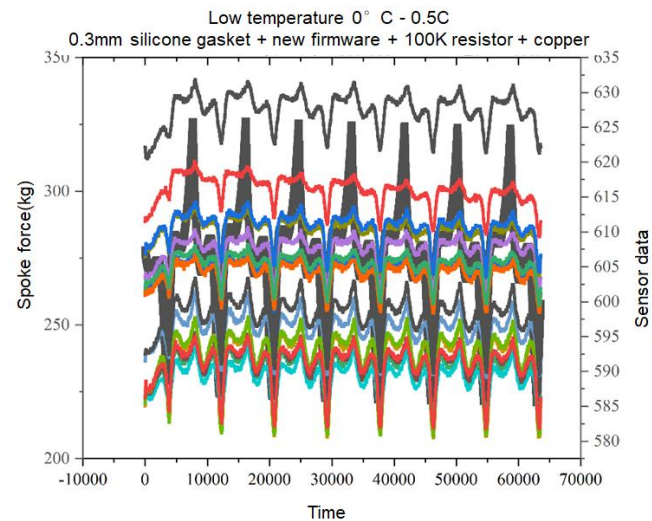
Thin-film sensing experiment at different temperatures within the module.



Thin-film sensing experiment at varying temperatures within the module.



The pressure change curve of the module at 50°C



The pressure change curve of the module at 0°C

Business Model

6

The patch-type MINI BSS system needs to be directly attached to the battery surface during the battery assembly process, so it is suitable for B2B cooperation with battery manufacturers.

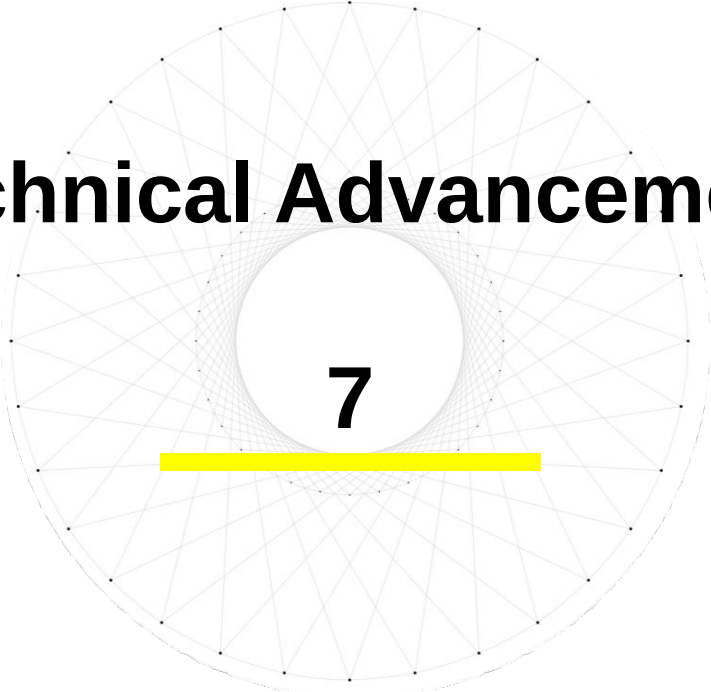
This model includes the following key points:

Partners: Directly cooperate with cell and battery pack manufacturers to integrate the BSS system into the production line to achieve early state monitoring and safety early warning.

Deep integration: Attach the Mini BSS system to the surface of the cell at the manufacturing stage of the cell and battery pack to monitor data such as voltage and temperature in real time and improve battery safety management.

Long-term cooperation: Usually through long-term agreements to ensure stable supply and provide reliable safety solutions for cell and battery pack manufacturers.

Technical Advancement



7

The MINI BSS system, as a world-first and innovative technology with completely independent intellectual property rights, can meet the global market's demand for efficient and safe battery management systems. Its advanced monitoring technology and intelligent chip solution have set a new industry benchmark.

Technological advancement and market competitiveness

Multi-physics field coupling monitoring technology

The BSS system integrates multi-physics field sensing technologies such as electricity, heat, and force, and can comprehensively evaluate the battery health status. It can effectively monitor battery pressure, impedance and temperature, and provide efficient early warning before thermal runaway, greatly enhancing safety.

Flexible thin film mechanical sensor

Ultra-thin flexible sensors are used to monitor the battery pressure distribution in real time, with high sensitivity and wide temperature adaptability (-20°C to 100°C), making it an ideal choice for precise battery management.

World's first battery passport and data encryption

Built-in battery passport and data encryption functions can effectively prevent illegal battery replacement, ensure data security throughout the entire life cycle, and enhance the safety and reliability of the supply chain and recycling links.

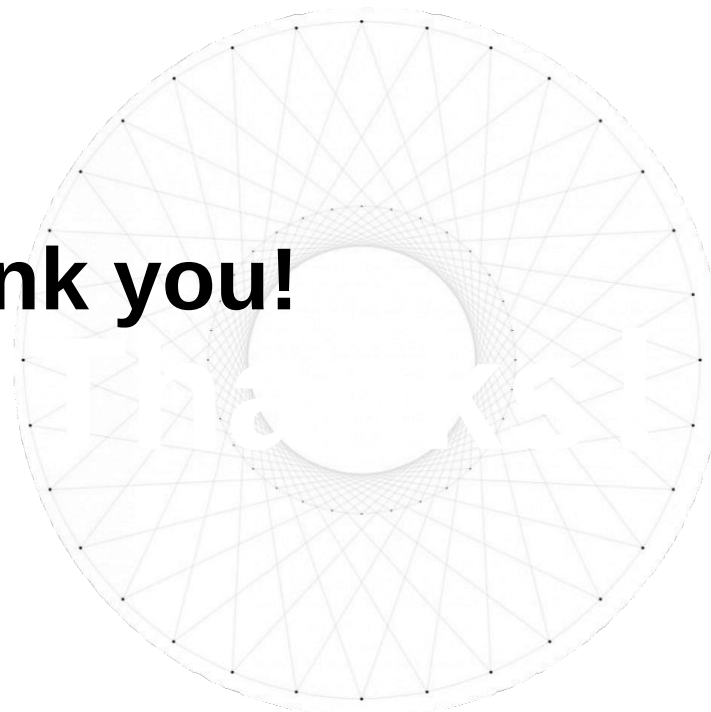
Multifunctional and highly integrated sensing chip

The world's first multifunctional chip supports temperature, voltage and pressure monitoring. The sensitivity is 120 times higher than traditional products and has edge AI computing capabilities to achieve rapid data analysis and relieve the pressure on data centers.

High cost performance and broad application prospects

The BSS system has high cost performance. With intelligent chips and integrated sensing technologies, it reduces costs and improves efficiency. It is suitable for large-scale applications in multiple industries such as electric vehicles and energy storage systems, and has broad market prospects.

Thank you!



Contact

Kretto China

Kevin Zheng

Kevin.zheng@kretto.com.cn

Mobile: +86 186 5326 9191